

1. At an auto repair shop, a routine maintenance is expected to take an average time of 50 minutes to complete, but it is suspected the actual time may be different. From a sample of 36 cars that receive maintenance, you find that the average time to complete it is 51.8 minutes. Assume the population standard deviation is known, $\sigma = 5.3$ minutes. Remember the hypothesis test framework: define hypotheses, collect data (and define variables), analyze data, state conclusions.
 - (a) Set up and list the hypotheses (H_0 and H_1) to test if the average number of minutes needed to complete the maintenance is **different** from 50, using a significance level of $\alpha = 0.05$.
 - (b) Evaluate the above hypothesis test (show all your calculations, including your test statistic (**z**) and its p-value) and state your conclusion in the context of the problem. (Do we reject or fail to reject H_0 , and what can we then say about the population mean?)
 - (c) Calculate an appropriately matching confidence interval (In other words, one with the same significance level α).
 - (d) Interpret your confidence interval (In other words, what can we say about our confidence that the true population mean in is this interval?).
 - (e) Does your hypothesis test conclusion match the conclusion from your confidence interval?
2. A small T-shirt company suspects their screen printer is producing too many defective shirts. The company expects 2% of products to have misprints (which are then discarded) but suspects that actual proportion may be **higher**. In the most recent print run of 400 T-shirts, 13 were misprinted.

Hints: To calculate σ (sigma), recall from class that the standard deviation σ of a Bernoulli random variable is $\sqrt{p * (1 - p)}$. When finding σ for the confidence interval, no p is provided, so you will need to use $\hat{p}$ instead. Note that this problem uses **proportions**, not means, and the parameter of interest is the **proportion** of misprints, NOT the number of misprints.

- (a) Set up hypotheses to test whether there is a **higher** proportion of misprints than expected, using a significance level of $\alpha = 0.01$.
- (b) Evaluate the above hypothesis test and state your conclusion.
- (c) Calculate an appropriately matching confidence interval.
- (d) Interpret your confidence interval.
- (e) Does your hypothesis test conclusion match the conclusion from your confidence interval?